Moment :

Let X be a RV and $C \in \mathbb{R}$ a scalar. Then, the kth moment of X is

$$E(X^k)$$

and k^{th} moment of X about C is

$$E[(X-C)^k].$$

We are interested in first moment of X : $\mu = E(X)$ and the second moment of X about μ : $Var(X) = E[(X-\mu)^2]$

Sample moment :

Let X be a RV, and $C \in \mathbb{R}$ a scalar. Let $x_1, x_2, \ldots, x_n$ be a sample from X.

The k^{th} sample moment of X is

$$\frac{1}{n} \sum_{i=1}^{n} x_i^k$$

The k^{th} sample moment of X (about c) is

$$\frac{1}{n} \sum_{i=1}^{n} (x_i - c)^k.$$

for eg: the first sample moment is just the sample mean and the second moment about mean is sample variance.

Method of Moments Estimation

Let $x_1, x_2, \ldots, x_n$ be sample observations from PMF $p_X(t;\theta)$ if X is discrete or PDF $f_X(t;\theta)$, if X is continuous, where θ is a parameter.

We define the method of moments (MoM) estimator $\hat{\theta}_{MoM}$ of $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ to be a solⁿ to the k simultaneous eqⁿs where, for $j = 1, 2, \dots, k$, we set jth true moment equal to jth sample moment:

$$E(X) = \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$$

$$\vdots$$

$$E(X^k) = \frac{1}{n} \sum_{i=1}^{n} x_i^k$$

① Let $x_1, x_2, \dots, x_n$ be ~~the~~ the sample observations from $X \sim Exp(\theta)$. What is the MoM estimator of θ?

Solⁿ: $k = 1$

$$E(X) = \frac{1}{\theta} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\Rightarrow \theta = \frac{n}{\sum_{i=1}^{n} x_i}$$

$$\Rightarrow \hat{\theta}_{MoM} = \frac{n}{\sum_{i=1}^{n} x_i}$$

② $X \sim Poi(\theta)$

Solⁿ: $k = 1$

$$E(X) = \theta = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\Rightarrow \hat{\theta}_{MoM} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$X \sim \text{Unif}(0, \theta)$

$$E(X) = \frac{\theta}{2} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\Rightarrow \theta = \frac{2}{n}\sum_{i=1}^{n} x_i$$

$$\Rightarrow \hat{\theta}_{MoM} = \frac{2}{n}\sum_{i=1}^{n} x_i$$

(4) $X \sim N(\theta_1, \theta_2)$, θ_1 - mean, θ_2 - variance.

Sol^n: $K = 2$

$$E(X) = \theta_1 = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\Rightarrow \hat{\theta}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$E(X^2) =$ $\left\{ Var(X) = E[X^2] - (E(X))^2 \right.$

$$\Rightarrow E(X^2) = Var(X) + (E(X))^2$$

$$\Rightarrow E(X^2) = \theta_2 + \theta_1^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$$

$$\Rightarrow \hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \hat{\theta}_1^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)^2$$